EcoBorder

The Free CBAM Toolkit

Complete User Guide

Every tool, every input & output, every calculation, and all references.

Live application: https://ecoborder.app

Estimation toolkit aligned to the EU Carbon Border Adjustment Mechanism (CBAM).

Contents

1. Introduction

1.1 What is CBAM?

The EU **Carbon Border Adjustment Mechanism (CBAM)** is a carbon tariff on imports of carbon-intensive goods — iron & steel, aluminium, cement, fertilisers, hydrogen and electricity. Importers must report the **embedded emissions** of these goods and, in the definitive regime (from 2026), surrender certificates covering them. Producers communicate their **Specific Embedded Emissions (SEE)** to importers using an official EU template.

1.2 What is EcoBorder?

EcoBorder is a free, browser-based toolkit that covers the whole CBAM chain: calculate embedded emissions from activity data, convert them to a carbon-border tax, look up CN codes, score readiness, and build the official installation communication. It is **tools-first** (no signup, no gating) and **private by design** — every calculation runs locally in the browser and nothing is sent to a server.

The CBAM chain

Activity data → Embedded emissions → Emission intensity (SEE) → CBAM tax → Official communication

1.3 Who is it for?

- Exporters/producers estimating their SEE and preparing the installation communication.
- Importers estimating CBAM liability and checking whether goods are in scope.
- Consultants and analysts who need fast, defensible estimates anchored to official EU data.

1.4 Baseline principle

CBAM is the single source of truth. All inputs, factors, vocabularies and outputs are anchored to the official EU CBAM regulation, default values, CN code list, production routes and the SEE communication template.

2. Quick Start

1. Open the app and go to the **Emissions Calculator**. Pick your product's CN code, enter production and your fuels/electricity. Read off the SEE intensity.
2. Click **Use in Tax Calculator** to estimate the CBAM liability, choosing your reporting year.
3. Use the **CN Code Checker** any time to confirm coverage and default factors.
4. When ready, click **Send to Communication Builder** (or open it directly) to assemble the official installation communication and export the Excel template.
5. New to it? On the Communication Builder, click **Load steel example** to see a fully-worked case.

3. Key Concepts & Glossary

Term	Meaning
Embedded emissions	Total greenhouse-gas emissions attributable to producing a good (tCO_2e).
SEE (Specific Embedded Emissions)	Embedded emissions per tonne of product (tCO_2e/t) — the emission intensity.
Direct emissions	Emissions from the production process itself (fuel combustion + process chemistry).
Indirect emissions	Emissions from the electricity consumed in production.
Precursor	An input material whose own embedded emissions cascade into the finished good (e.g. crude steel into steel products). May be produced internally or purchased.
Production route	The method used to make a good (e.g. Blast furnace, Basic oxygen steelmaking, Electric arc furnace).
Aggregated goods category	The CBAM grouping a CN code belongs to (e.g. Crude steel, Pig iron, Cement).
CN code	8-digit Combined Nomenclature code identifying the exact product.
CBAM factor	Share of embedded emissions for which certificates are due in a given year (free allocation phases out: 2.5% in 2026 to 100% in 2034).
Free allocation	The exempted share (1 - CBAM factor) during the phase-in.
Carbon price paid at origin	Carbon price already paid in the producing country, credited against CBAM.
Reporting period	Annual basis for the installation communication.

4. Tool 1 — Emissions Calculator

Purpose: compute bottom-up embedded emissions and the SEE intensity of a product from real activity data, following the CBAM SEE methodology.

4.1 Inputs (field by field)

Section 1 — Product & reporting period

- **CN code (official):** search and select the exact 8-digit code. This drives the aggregated goods category and the list of relevant precursors.
- **Product name:** free text for your reference.
- **Sector / Aggregated goods category:** auto-set from the CN code; editable.
- **Production route:** official options for the category (e.g. Blast furnace, EAF).
- **Annual production (tonnes):** the SEE denominator.
- **Reporting year:** annual basis (no quarter).

Section 2 — Direct emissions (Scope 1)

- **Fuel combustion:** add fuels burnt at your installation (type x quantity in native units).
- **Precursor inputs:** auto-listed from the CN category. Enter the amount used; emissions are computed from each precursor's official SEE.
- **Process emissions:** measured non-combustion chemistry (calcination, reduction, PFCs) entered explicitly.
- **Measurable heat & waste gas:** imported/exported energy (TJ) x emission factor.

Section 3 — Indirect emissions (Scope 2)

- **Electricity consumed (MWh) x grid emission factor**, minus exported electricity.

> **Avoiding double counting**
>
> Fuels represent combustion at **your** installation. A purchased precursor's factor already includes the energy used to make it — so do not also enter that fuel. The two are kept in separate sub-sections with an on-screen warning.

4.2 Calculation

```
fuelDirect    = Σ (fuel quantity × fuel emission factor)
precDirect    = Σ (precursor amount × precursor SEE_direct)
precIndirect = Σ (precursor amount × precursor SEE_indirect)
heatAdj       = (heat imported − heat exported) × heat EF
wasteGasAdj   = (WG imported − WG exported) × WG EF
ownDirect     = fuelDirect + processEmissions + heatAdj + wasteGasAdj
```

```
direct       = ownDirect + precDirect
ownIndirect  = (electricity MWh × grid EF) − (exported MWh × export EF)
indirect     = ownIndirect + precIndirect
SEE direct   = direct ÷ production
SEE indirect = indirect ÷ production
intensity    = (direct + indirect) ÷ production        (all to 2 dp)
```

4.3 Worked example (steel, blast furnace)

Inputs

Production = 4,800,000 t · Directly attributable emissions = 7,866,044 tCO_2e · Waste gas exported = 12,800 TJ at 37.4187 tCO_2/TJ · Electricity = 1,658,844 MWh at 0.589 tCO_2/MWh

```
wasteGasAdj = (0 − 12,800) × 37.4187   = −478,959 tCO2e
direct      = 7,866,044 − 478,959      = 7,387,085 tCO2e
indirect    = 1,658,844 × 0.589        =   977,059 tCO2e
SEE direct  = 7,387,085 ÷ 4,800,000    = 1.54 tCO2e/t
SEE indirect= 977,059 ÷ 4,800,000      = 0.20 tCO2e/t
intensity   = 1.54 + 0.20              = 1.74 tCO2e/t
```

This matches the official EU example exactly (see Section 11).

4.4 Outputs

- SEE direct, SEE indirect, total embedded emissions, intensity (tCO_2e/t), indicative CBAM cost — all to two decimals.
- Hand-offs: **Use in Tax Calculator** and **Send to Communication Builder**; plus JSON export.

5. Tool 2 — CBAM Tax Calculator

Purpose: estimate CBAM certificate liability across product lines, applying the year-based phase-in, free allocation and any carbon price paid at origin.

5.1 Inputs

- **Reporting year** (sets the CBAM phase-in factor) and **EU ETS reference price** (default 82 €/tCO2).
- Per line: product, sector, annual volume (t), **emission intensity** (tCO2/t, usually imported from the Emissions Calculator), and **carbon price already paid at origin** (€/tCO2).

5.2 Calculation

```
embedded     = volume × intensity
factor       = CBAM phase-in factor for the year   (taxable share)
chargeable   = embedded × factor
gross        = embedded × ETS price
freeRelief   = gross × (1 − factor)                (exempt: free allocation)
originCredit = chargeable × min(origin price, ETS price)
net          = max(gross − freeRelief − originCredit, 0)
quarterly    = net ÷ 4
```

5.3 Threshold logic — taxable vs exempted

There is **no emissions-quantity threshold**. The **taxable** portion is the embedded emissions multiplied by the CBAM phase-in factor for the year. The **exempted** portion is the free-allocation share (1 - factor), which declines each year. Any carbon price already paid at origin is credited against the chargeable portion. (CBAM's only customs exemption is the EUR 150 per-consignment rule, which is not modelled.)

5.4 CBAM phase-in factor schedule

Year	CBAM factor (taxable)	Free allocation (exempt)
2026	2.5%	97.5%
2027	5%	95%
2028	10%	90%
2029	22.5%	77.5%
2030	28.5%	71.5%
2031	48.75%	51.25%

Year	CBAM factor (taxable)	Free allocation (exempt)
2032	63.75%	36.25%
2033	86%	14%
2034	100%	0%

Later-year values remain subject to EU confirmation and are editable in the tool.

5.5 Worked example

1,000 t of steel at intensity 1.74, reporting year 2027 (factor 5%), ETS 82 €/t, no origin price

```
embedded   = 1,000 × 1.74 = 1,740 tCO2e
chargeable = 1,740 × 0.05 = 87 tCO2e
gross      = 1,740 × 82   = €142,680
freeRelief = 142,680 × 0.95 = €135,546
net        = 142,680 − 135,546 = €7,134   (= 87 × 82)
```

5.6 Outputs

- Per line: embedded, chargeable, relief + credits, net liability; liability-by-line chart; total; indicative quarterly figure.
- CSV and JSON export.

6. Tool 3 — CN Code Checker

Purpose: determine whether a product is covered by CBAM and retrieve its official classification and default emission factors.

6.1 Inputs & data

- Free-text search (code, product name or category) and a sector filter.
- **749 official CBAM CN codes** parsed from the EU communication template — exact code, CN name and aggregated goods category.
- Official transitional **direct and indirect default factors** at aggregated-category level.

6.2 Calculation

```
total          = direct + indirect
cost per tonne = total × ETS price (82)
annual (1,000t)= total × 1,000 × 82
```

Electricity is flagged as **country-specific** (IEA factor), not a single tCO2/t default.

6.3 Outputs

- Filtered list plus a detail panel: category, direct/indirect/total factors, per-tonne and 1,000-tonne cost, and a link to the Tax Calculator.

7. Tool 4 — Readiness Check

Purpose: a two-minute self-assessment of CBAM preparedness with a concrete action list.

7.1 Inputs & scoring

- Five questions (exports in scope, CN classification, emissions measurement, supplier data, reporting), each scored 0-2.
- Score% = (sum of answers ÷ 10) × 100.
- Tiers: 75%+ 'CBAM Ready'; 40-74% 'Getting There'; below 40% 'At Risk'.

7.2 Outputs

- A score gauge and a prioritised action list generated from the gaps (questions below maximum). Shown free, no email.

8. Tool 5 — CBAM Communication Builder

Purpose: assemble the official installation SEE communication (the document a producer gives EU importers) and export it in the official template structure.

8.1 The five steps & their fields

Step 1 — Installation

- Name (local & English), economic activity, full address, country (origin), UNLOCODE, coordinates, authorised representative; annual reporting period.
- Optional **verifier & accreditation** block (company, address, representative, accreditation body and registration number).

Step 2 — Production processes

- Aggregated goods category, **official production route**, total production, directly attributable emissions.
- Measurable heat & waste-gas import/export; electricity & electricity export.
- **Internal precursors**: consume the output of another process within the installation (e.g. pig iron feeding crude steel).

Step 3 — Purchased precursors

- Country, route, quantity and SEE — cascaded into the consuming process.

Step 4 — Products

- Official CN code (auto name & category), product name for invoices, SEE direct/indirect (prefilled from the process, editable), share-by-default.
- **Sector-specific properties**: steel (% Mn/Cr/Ni/C, scrap ratio, mill ID), aluminium (scrap, % non-Al), cement (clinker factor, calcined), fertilisers (% N/ammonium/nitrate/urea).
- **Carbon price paid at origin** (instrument, currency, CP due, rebate -> effective CP).

Step 5 — Export

- Review and download.

8.2 Calculation (SEE resolver)

```
ownDirect(process)   = directEmissions + heatAdj + wasteGasAdj
```

```
ownIndirect(process) = (electricity × EF) − (exported × EF)
```

```
direct   = ownDirect   + Σ purchased(qty × SEE_d) + Σ internal(qty × upstreamSEE_d)
```

```
indirect = ownIndirect + Σ purchased(qty × SEE_i) + Σ internal(qty × upstreamSEE_i)
```

```
SEE direct = direct ÷ production    SEE indirect = indirect ÷ production
```

Internal precursor chains (sintered ore -> pig iron -> crude steel -> steel products) are resolved recursively with a cycle guard. Products inherit their process SEE. Effective carbon price = max(CP due - rebate, 0).

8.3 Export format (official template structure)

Excel tab	Contents
Installation	Reporting period, installation details, verifier & accreditation, total direct/indirect/total emissions.
Summary_Products	Per product: process, route, type, CN code, CN name, product name, SEE direct/indirect/total, unit, share-by-default, electricity EF, properties, carbon price & effective CP.
Communication	Importer-facing summary: installation, totals, and the per-product SEE table.

- Also exports JSON and supports Print. **Load steel example** pre-fills the validated blast-furnace case.

9. Consolidated Calculation Reference

Emissions / SEE

```
direct   = Σ(fuel×EF) + processEmissions + (heatImp−heatExp)×heatEF
           + (wgImp−wgExp)×wgEF + Σ(precursor amt × SEE_direct)
indirect = (elec MWh × gridEF) − (export MWh × exportEF) + Σ(precursor amt × SEE_indirect)
SEE_direct = direct ÷ production    SEE_indirect = indirect ÷ production
```

CBAM tax

```
embedded    = volume × intensity
net         = embedded × ETSprice × factor(year) − embedded×factor×min(origin,ETS)
            = chargeable × (ETSprice − min(origin,ETS))
```

Carbon price (Communication Builder)

```
effectiveCP = max(CP_due − rebate, 0)
```

10. Data Sources & References

Data	Source	Status
CN-code default factors (direct/indirect)	EU Commission, 'Default values for the transitional period of the CBAM', 22 Dec 2023	Official
749 CN codes, names, categories	Official EU 'CBAM Communication template for installations' (V2, 2024)	Official
Production routes & precursor relationships	Same official template (Parameters)	Official
CBAM phase-in factor schedule	EU free-allocation phase-out (2.5% 2026 to 100% 2034)	Official*
Fuel & grid emission factors	IPCC / IEA-style reference values	Indicative
EU ETS reference price (82 €/tCO2)	Reference price for valuation	Editable

* Later-year factors remain subject to EU confirmation. Definitive-regime (2026+) default values are country-specific (Implementing Regulation (EU) 2025/2621) and are not yet encoded; the transitional EU-wide values are used.

Primary reference document: EU CBAM transitional default values (PDF).

11. Validation

The engine was validated against the official filled example 'CBAM SEE V2.1 - Example Steel 1 (Blast furnace)'.

Result	EcoBorder	Official workbook
SEE direct	1.54 (1.5390)	1.53898
SEE indirect	0.20 (0.2036)	0.20355
SEE total	1.74 (1.7425)	1.74253

The figures match exactly; values are displayed to two decimals per requirement.

12. FAQ

Do I need to sign up or share data?

No. Every tool is free and runs entirely in your browser; nothing you enter is sent to a server.

Are the numbers official?

CN-code default values and the CN catalogue are official EU data. Fuel and grid factors in the Emissions Calculator are indicative IPCC/IEA-style values for estimation.

Annual or quarterly reporting?

The installation SEE communication is annual. The reporting year drives the CBAM phase-in factor in the Tax Calculator.

How are precursors handled?

Relevant precursors are auto-derived from the CN code's category. Internally-produced precursors are modelled as linked processes; purchased ones carry their own SEE. Both cascade into the product, with double-count protection against fuels.

Can I export an official submission file?

You can export an Excel workbook mirroring the official template's structure. Validate it against the current EU template/registry before any real submission; a schema-valid registry XML requires the official EU XSD (a planned next step).

13. Scope & Caveats

- EcoBorder is an estimation and planning toolkit - **not** legal, financial or compliance advice.
- CN-code default values are official; fuel/grid factors are indicative.
- The Excel export mirrors the official template's structure and columns (a clean workbook), not the protected original with embedded formulas.
- Always confirm figures and obligations against current official EU CBAM guidance before filing.